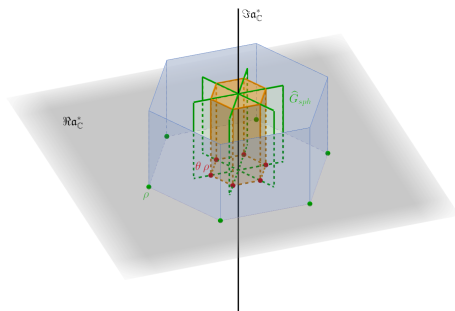


Spectral theory of automorphic forms

Chris Lutsko (U Houston)



Plan for talk:

I. Spectral and Patterson-Sullivan theory of $L^2(\Gamma \backslash \mathbb{H})$

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- II. Apollonian circle packings
- III. Higher rank spectral theory

The upper half-plane

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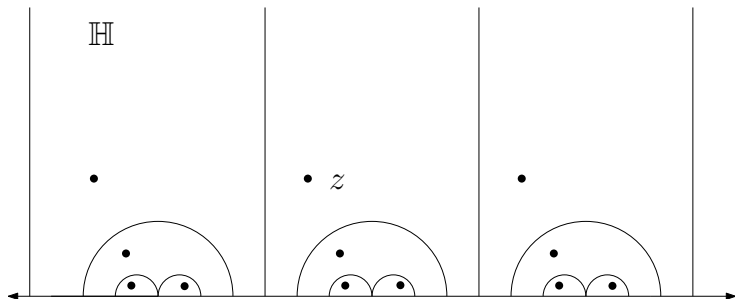
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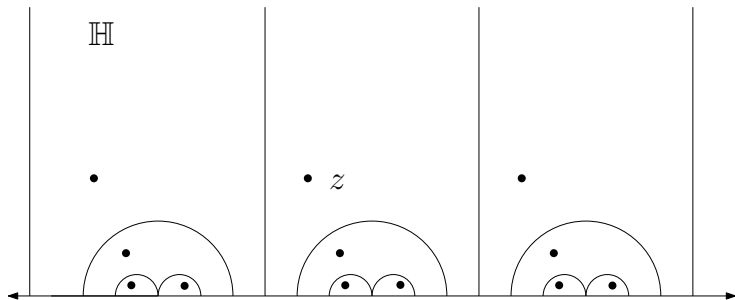
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Example of a non-lattice:

$$\Gamma_\mu := \left\langle \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix} \right\rangle$$

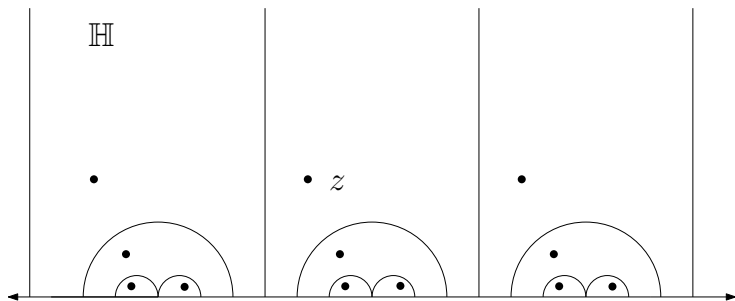


Definitions



$\Lambda(\Gamma)$ - **limit set**: the set of accumulation points of an orbit. $\mathbb{R}$ for lattices, fractal for non-lattices.

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δ_Γ - **critical exponent**: the exponential growth rate of points

$$\delta_\Gamma := \inf \left(s : \sum_{\gamma \in \Gamma} e^{-sd(z, \gamma z)} < \infty \right)$$

for any $z \in \mathbb{H}$. Lattices: $\delta_\Gamma = 1$, non-lattices: $\delta_\Gamma < 1$.

Consider the hyperbolic Laplacian

$$\Delta := -y^2(\partial_{xx} + \partial_{yy})$$

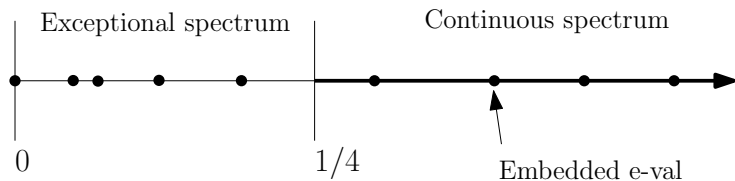
and let $\sigma(\Delta) := \text{Spec}_{L^2(\Gamma \backslash G)}(\Delta)$.

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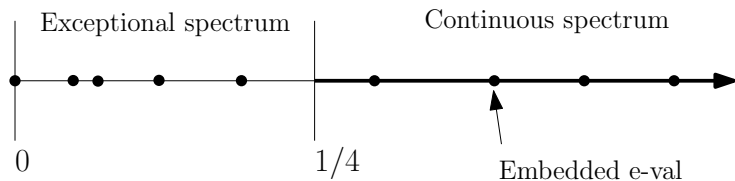


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Given a $\lambda \in \sigma(\Delta)$ we write $\lambda = s(1 - s)$ for $s \in \mathbb{C}$.

Connecting spectrum to critical exponent

Theorem (Elstrodt (1973), Patterson (1976))

When $\Gamma < \mathrm{SL}_2(\mathbb{R})$ is a discrete, non-lattice subgroup, then

$$\inf(\sigma(\Delta)) = \begin{cases} 1/4 & \text{if } \delta_\Gamma < 1/2 \\ \delta_\Gamma(1 - \delta_\Gamma) & \text{if } \delta_\Gamma \geq 1/2. \end{cases}$$

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Lax-Phillips (1983) then show there are no embedded eigenvalues.

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Warning: This theorem is not in general true, even in rank 1 (see Dufloux 2017).

Matrix coefficients

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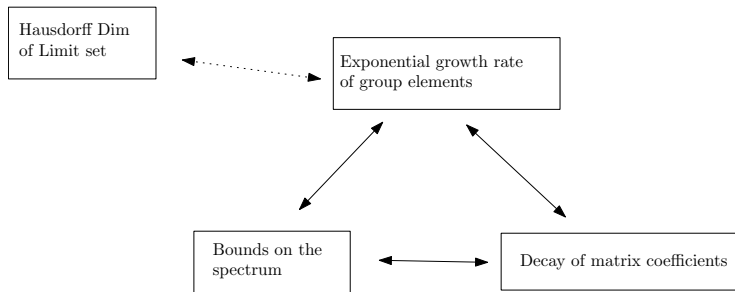
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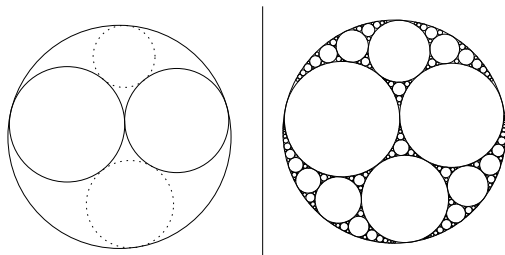
In fact one can quantify the decay of matrix coefficients when $\delta_\Gamma > 1/2$.

Upshot: Trichotomy



II: Apollonian counting problem

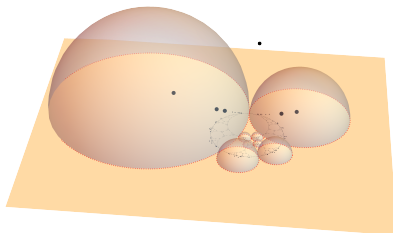
The image below is an Apollonian circle packing:



Apollonian counting problem: Let $N_P(T)$ denote the number of circles of radius $> 1/T$. Can we estimate $N_P(T)$?

II: Apollonian counting problem

The packing can be embedded in $\partial\mathbb{H}^3$, with generating group Γ for which the ACP is the limit set



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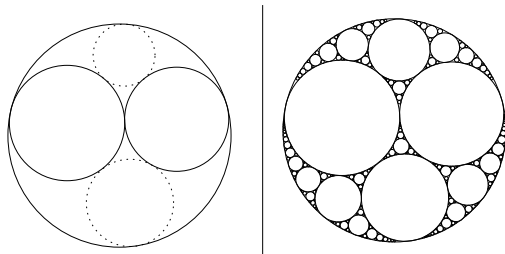
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Maximal spectral gap (of size ≈ 0.093) (Kelmer-Kontorovich-L).



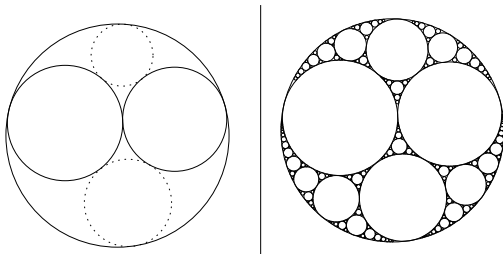
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$$N_P(T) = c_P T^{\delta_\Gamma} + O(T^{\delta_\Gamma - \eta}) \quad (1)$$

$$\eta = \frac{2}{5}(\delta_\Gamma - 1) \approx 0.12 \text{ (Kontorovich-L).}$$

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Building on Boyd, Kontorovich-Oh, Lee-Oh, Vinogradov.

III: Higher rank setup

Let G be a real, connected, semi-simple Lie group with finite center, rank r and maximal compact, K . Then G admits a Cartan decomposition

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$$\psi_\Gamma(H) := \|H\| \inf_{H \in \mathcal{C}} \inf \left\{ s \in \mathbb{R} \mid \sum_{\gamma \in \Gamma, \mu_+(\gamma) \in \mathcal{C}} e^{-s\|\mu_+(\gamma)\|} < \infty \right\},$$

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where the infimum runs over all open cones $\mathcal{C} \subseteq \mathfrak{a}$ with $H \in \mathcal{C}$.

III: Spectral theory of ring of invariant differential operators

Rather than work with just the Laplacian, we work with the full algebra of invariant differential operators on G/K which we denote by $\mathbb{D}(G/K)$.

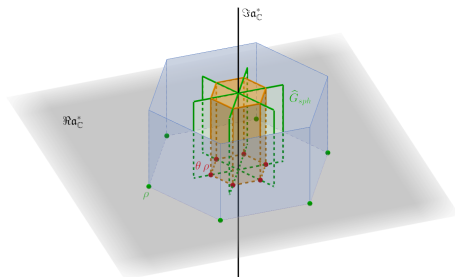
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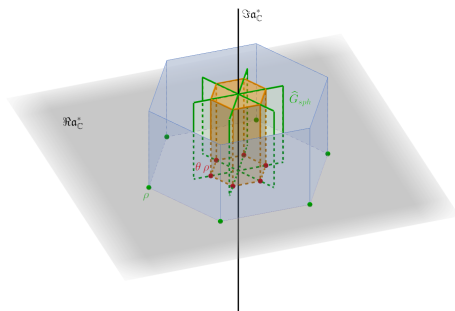
Note that $\Re \tilde{\sigma} \subseteq \text{conv}(W\rho)$, where ρ denotes the half-sum of restricted roots and $\text{conv}(W\rho)$ is the polyhedron described by the convex hull of the Weyl orbit of ρ .



III: Polyhedral norm

To quantify how far the spectrum lies from the boundary of $\text{conv}(W\rho)$, we introduce the following norm: For any homogeneous (not necessarily linear) function $\lambda : \mathfrak{a} \rightarrow \mathbb{R}$

$$\|\lambda\|_{poly} = \sup_{w \in W, H \in \overline{\mathfrak{a}_+}} \frac{\lambda(wH)}{\rho(H)}.$$



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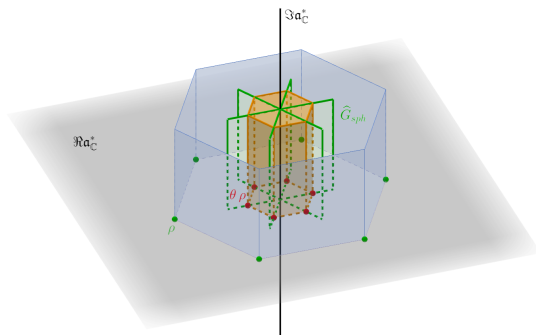
for some $C > 0$ independent of the choice of v or test functions f_1, f_2 .

IV: Main theorem

Theorem (L., Weich, Wolf 2024)

Let G be a real semisimple connected non-compact Lie group with finite center and $\Gamma < G$ a discrete and torsion-free subgroup. Then

$$\sup_{\lambda \in \tilde{\sigma}} \|\Re \lambda\|_{poly} = \max(0, \|\psi_\Gamma - \rho\|_{poly}) = \theta(L^2(\Gamma \backslash G)). \quad (2)$$



IV: Corollary – Temperedness

Corollary (L., Weich, Wolf 2024)

$L^2(\Gamma \backslash G)$ is tempered if and only if $\psi_\Gamma \leq \rho$ which occurs if and only if $\tilde{\sigma} \subset \mathfrak{S}\mathfrak{a}_\mathbb{C}^$.*

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This confirms a conjecture of Hee Oh.

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Dey, Kim, Oh (2024): For a $P_{\min}$ -Anosov subgroup, they bound the growth indicator function in terms of the Hausdorff dimension of the limit set in $G/P_{\min}$. (Extended to general P with Theorem LWW)

V: Preparations to ideas in proof

For any $\lambda \in \mathfrak{a}_{\mathbb{C}}^*$ we can define the *elementary spherical function*

$$\phi_{\lambda}(g) := \int_K e^{-(\lambda+\rho)H(g^{-1}k)} dk$$

where $H(g)$ denotes the A -component in KAN .

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$$D\phi_{\lambda} = \chi_{\lambda}(D)\phi_{\lambda} \quad \forall D \in \mathbb{D}(G/K).$$

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There is a constant C such that

$$e^{-\rho(H)} \leq \Xi(e^H) \leq C(1 + |H|)^d e^{-\rho(H)} \quad (3)$$

for $H \in \mathfrak{a}_+$. Here d is the number of positive reduced roots.

V: Irreducible representations

By the abstract Plancherel theory, the right regular representation can be decomposed into a direct integral of irreducible representations

$$(R, L^2(\Gamma \backslash G)) \simeq \int_{\widehat{G}}^{\oplus} \pi_x \, d\mu(x)$$

where $\widehat{G}$ is the unitary dual.

V: Relating spectrum to decay estimates for m.c's

Goal: $\forall \varepsilon > 0, \exists d_\varepsilon > 0$ s.t. $\forall f, g \in L^2(\Gamma \backslash G)^K$

$$|\langle R(\exp v)f, g \rangle| \leq d_\varepsilon e^{\sup_{\lambda \in \tilde{\sigma}} (\Re \lambda - \rho)(v)} e^{\varepsilon \|v\|} \|f\|_2 \|g\|_2.$$

Proof: Decompose f, g into $\int_X^\oplus f_x d\mu(x)$ and $\int_X^\oplus g_x d\mu(x)$. Since f, g are K invariant, we have

$$\begin{aligned} \langle R(\exp v)f, g \rangle &= \int_{\pi^{-1}(\widehat{G}_{sph})} \phi_{\lambda_x}(\exp v) \langle f_x, g_x \rangle d\mu(x) \\ &\leq \int_{\pi^{-1}(\widehat{G}_{sph})} |\phi_{\lambda_x}(\exp v)| \|f_x\| \|g_x\| d\mu(x) \\ &\leq \operatorname{esssup}_{\pi_* \mu|_{\widehat{G}_{sph}}} |\phi_{\lambda_x}(\exp v)| \|f\|_2 \|g\|_2 \\ &\leq \sup_{\lambda \in \tilde{\sigma}} |\phi_\lambda(\exp v)| \|f\|_2 \|g\|_2. \end{aligned}$$

For the elementary spherical function we have the well-known estimates

$$|\phi_\lambda(\exp v)| \leq e^{\Re\lambda(v)} \Xi(\exp v) \leq d_\varepsilon e^{\Re\lambda(v)} e^{-\rho(v)} e^{\varepsilon\|v\|}$$

for $\Re\lambda \in \overline{\mathfrak{a}_+^*}$ and any $\varepsilon > 0$.

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V:Sketch proof cont.

Furthermore, we require the following analogue of a wave-front lemma. Recall the Bruhat decomposition $\overline{N} \times M \times A \times N \subset G$ (full measure open submanifold). Moreover

$$\int_G f(g) dg = \int_{\overline{N} \times M \times A \times N} f(\overline{n}man) e^{2\rho(\log a)} d\overline{n} dm da dn$$

Lemma

Let $\varphi_1, \varphi_2 \in C_c(G)$. Then there exists a constant C such that for all $h \in A$

$$\left| \int_G \varphi_1(h^{-1}gh) \varphi_2(g) dg \right| \leq C e^{-2\rho(\log h)} \quad (4)$$

V: Growth indicator bounds imply decay of m.c

Theorem

Let $f_1, f_2 \in C_c(\Gamma \backslash G)$, $H_0 \in \overline{\mathfrak{a}_+}$ normalized, and $s > \psi_\Gamma(H_0)$, $s \geq 0$. Then there exists $\delta > 0$ and $C > 0$ such that

$$|\langle R(\exp tH)f_1, f_2 \rangle_{L^2(\Gamma \backslash G)}| \leq Ce^{t(s-2\rho(H))}$$

for all $t \geq 0$ and $H \in B_\delta(H_0)$ normalized.

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We can find $\tilde{f}_i \in C_c(G)$ such that $f_i(\Gamma g) = \sum_{\gamma \in \Gamma} \tilde{f}_i(\gamma g)$.

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We can find $\tilde{f}_i \in C_c(G)$ such that $f_i(\Gamma g) = \sum_{\gamma \in \Gamma} \tilde{f}_i(\gamma g)$.

$$\begin{aligned} \langle R(h) f_1, f_2 \rangle_{L^2(\Gamma \backslash G)} &= \int_{\Gamma \backslash G} f_1(\Gamma gh) f_2(\Gamma g) d\Gamma g = \int_G \tilde{f}_1(gh) f_2(\Gamma g) dg \\ &= \sum_{\gamma \in \Gamma} \int_G \tilde{f}_1(gh) \tilde{f}_2(\gamma g) dg. \end{aligned} \tag{5}$$

V: Growth indicator bounds imply decay of m.c

If we define $\varphi_i(g) := \max_{s \in \text{supp } \tilde{f}_i} |\tilde{f}_i(sg)|$ we can estimate

$$\left| \int_G \tilde{f}_1(gh) \tilde{f}_2(\gamma g) dg \right| \leq \int_G \varphi_1(h^{-1}gh) \varphi_2(g) dg.$$

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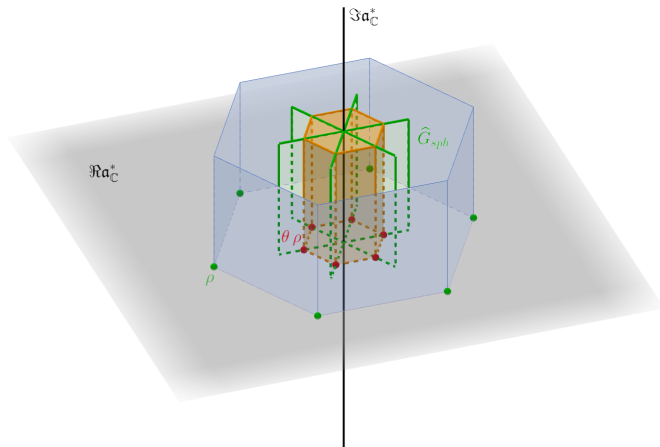
$$\left| \int_G \tilde{f}_1(gh) \tilde{f}_2(\gamma g) dg \right| \leq \int_G \varphi_1(h^{-1}gh) \varphi_2(g) dg.$$

Hence we have

$$|\langle R(h)f_1, f_2 \rangle| \leq \#(\Gamma \cap (\text{supp } \tilde{f}_2)h(\text{supp } \tilde{f}_1)^{-1}) \int_G \varphi_1(h^{-1}gh) \varphi_2(g) dg.$$

Apply lemma

For $G = \mathrm{SL}_3(\mathbb{R})$ we actually have $\sup_{\lambda \in \tilde{\sigma}} \|\Re \lambda\|_{\mathrm{poly}} = \theta$



THANK YOU!!!